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Camera calibration and distance measurement based on homography

Abstract

Example implementations include a method, apparatus and computer-readable medium for estimating a real world height in an image, comprising obtaining two images of an object, with known height, wherein each image depicts the object at a different position that is estimated using a homography matrix. The homography matrix transforms an image plane associated with the image to a real world plane. The implementations include generating a first vector that intersects with the object and points to a real world ground plane using the first image and a second vector that intersects with the same point on the object and points to the real world ground plane using the second image. Additionally, the implementations further include calibrating the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/384,847, filed Nov. 23, 2022, which is herein incorporated by reference.

BACKGROUND

Technical Field

(1) The described aspects relate to camera and computer vision systems.

Introduction

(2) Height estimation is useful in many world applications. For example, the heights of vehicles may be estimated for when height restrictions exist (e.g., tunnels, bridges, etc.). In another example, height may be used by security applications to distinguish children from adults. In terms of computer vision, estimating object height from a single image requires camera calibration parameters (both intrinsic and extrinsic matrices). Camera calibration is the process of determining camera parameters in order to complete operations with specified performance measurements. Unfortunately, the calibration process is often complicated and not always possible on the field.

SUMMARY

(3) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(4) An example aspect includes a method for estimating a real world distance in an image, comprising obtaining a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height. The method further includes estimating, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane. Additionally, the method further includes determining, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground

plane. Additionally, the method further includes obtaining a second image from the camera, wherein the second image includes the first object at a second location in the environment. Additionally, the method further includes estimating, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane. Additionally, the method further includes determining, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the second real world ground plane. Additionally, the method further includes calibrating the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

(5) Another example aspect includes an apparatus for estimating a real world distance in an image, comprising one or more memories and one or more processors coupled with the one or more memories. The one or more processors individual or in combination are configured to obtain a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height. The one or more processors individually or in combination are further configured to estimate, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane. Additionally, the processor further configured to determine, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane. Additionally, the processor further configured to obtain a second image from the camera, wherein the second image includes the first object at a second location in the environment. Additionally, the processor further configured to estimate, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane. Additionally, the processor further configured to determine, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane. Additionally, the processor further configured to calibrate the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

(6) Another example aspect includes an apparatus for estimating a real world distance in an image, comprising means for obtaining a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height. The apparatus further includes means for estimating, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane. Additionally, the apparatus further includes means for determining, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane. Additionally, the apparatus further includes means for obtaining a second image from the camera, wherein the second image includes the first object at a second location in the environment. Additionally, the apparatus further includes means for estimating, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane. Additionally, the apparatus further includes means for determining, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the second real world ground plane. Additionally, the apparatus further includes means for calibrating the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

(7) Another example aspect includes a computer-readable medium having instructions stored thereon for estimating a real world distance in an image, wherein the instructions are executable by a processor to obtain a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first

object has a known object height. The instructions are further executable to estimate, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane. Additionally, the instructions are further executable to determine, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane. Additionally, the instructions are further executable to obtain a second image from the camera, wherein the second image includes the first object at a second location in the environment. Additionally, the instructions are further executable to estimate, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane. Additionally, the instructions are further executable to determine, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the second real world ground plane. Additionally, the instructions are further executable to calibrate the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

(8) An example aspect includes a method for estimating a real world distance in an image, comprising obtaining an image from a camera installed in an environment. The method further includes identifying, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane. Additionally, the method further includes determining a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines. Additionally, the method further includes identifying a ground line connecting the first intersection point and the second intersection point. Additionally, the method further includes estimating an internal camera matrix based on a focal length and an image size associated with the camera. Additionally, the method further includes calibrating the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the internal camera matrix.

(9) Another example aspect includes an apparatus for estimating a real world distance in an image, comprising a memory and a processor coupled with the memory. The one or more processors individually or in combination are configured to obtain an image from a camera installed in an environment. The one or more processors individually or in combination are further configured to identify, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane. Additionally, the processor further configured to determine a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines. Additionally, the processor further configured to identify a ground line connecting the first intersection point and the second intersection point. Additionally, the processor further configured to estimate an internal camera matrix based on a focal length and an image size associated with the camera. Additionally, the processor further configured to calibrate the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the internal camera matrix.

(10) Another example aspect includes an apparatus for estimating a real world distance in an image, comprising means for obtaining an image from a camera installed in an environment. The apparatus further includes means for identifying, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane. Additionally, the apparatus further includes means for determining a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines. Additionally, the apparatus further includes means for identifying a ground line connecting the first intersection point and the second intersection point. Additionally, the apparatus further includes means for estimating an internal camera matrix based on a focal length and an image size associated with the camera. Additionally, the apparatus further includes means for calibrating the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the

internal camera matrix.

(11) Another example aspect includes a computer-readable medium having instructions stored thereon for estimating a real world distance in an image, wherein the instructions are executable by a processor to obtain an image from a camera installed in an environment. The instructions are further executable to identify, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane. Additionally, the instructions are further executable to determine a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines. Additionally, the instructions are further executable to identify a ground line connecting the first intersection point and the second intersection point. Additionally, the instructions are further executable to estimate an internal camera matrix based on a focal length and an image size associated with the camera. Additionally, the instructions are further executable to calibrate the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the internal camera matrix.

(12) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements, wherein dashed lines may indicate optional elements, and in which:

(2) FIG. 1A is an image of an object captured by a camera to be calibrated;

(3) FIG. 1B is a second image of the object in a different position;

(4) FIG. 2 is a diagram showing points on an image plane being mapped to points on a ground plane using a homography matrix;

(5) FIG. 3A is a graph of a z-axis distance metric versus an y-axis distance metric, in real world distances, of a first vector that intersects with a point on the object (in FIG. 1) and a ground plane;

(6) FIG. 3B is a graph of a z-axis distance metric versus an y-axis distance metric, in real world distances, of an intersection between the first vector (in FIG. 3A) and a second vector drawn based on the object at a different position;

(7) FIG. 4 is an image of an object on a surface with parallel lines as captured by a camera to be calibrated, and as projected onto a graph of a z-axis distance metric versus an y-axis distance metric, in real world distances;

(8) FIG. 5 is a diagram of a detection plane associated with the camera calibrated based on the image of FIG. 4;

(9) FIG. 6 is a block diagram of an example of a computer device having components configured to perform a method for estimating a real world distance in an image;

(10) FIG. 7 is a flowchart of an example of a method for estimating a real world distance in an image;

(11) FIG. 8 is a flowchart of additional aspects of the method of FIG. 7;

(12) FIG. 9 is a block diagram of an example of a computer device having components configured to perform a method for estimating a real world distance in an image;

(13) FIG. 10 is a flowchart of an example of a method for estimating a real world distance in an image; and

(14) FIG. 11 is a flowchart of additional aspects of the method of FIG. 10.

DETAILED DESCRIPTION

(15) Various aspects are now described with reference to the drawings. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of one or more aspects. It may be evident, however, that such aspect(s) may be practiced without these specific details.

(16) The present disclosure describes systems and methods to calibrate a camera and to measure real world distances, such as height, in an image captured by the camera. In one approach, the calibration for subsequent measurement is performed using image to ground homography. In another approach, the calibration for subsequent measurement is performed using parallel lines on the ground in the image, along with a camera focal length and an object of known height in the image. In either approach, the calibrated camera can then be used to estimate a height and/or size of other objects in an image captured by the camera. In general, the approaches may be used to estimate any distance in a plane perpendicular to the plane on which the homography is calculated (e.g., the ground plane). Conventional cameras are unable to estimate height using a single image due to a lack of proper calibration. Achieving the proper calibration, as specifically described in the present disclosure, can thus improve camera technology and enable accurate height estimations.

(17) The described aspects provide a simple and efficient technique, as compared to prior complicated and intensive techniques, for calibrating a camera to determine real world measurements. For instance, estimating object height from a single image requires full camera calibration parameters (both intrinsic and extrinsic matrices). Such a full calibration process is complicated and not always possible in the field. The described aspects include two approaches for height estimation with two calibration processes. The calibration processes provide sufficient calibration parameters to provide accurate estimations that can be used to estimate object measurements. The described aspects may be useful in many practical applications. For instance, height estimation from a security camera might be useful in many world applications, including estimating heights for vehicles when height restriction exists, such as in the case of vehicles going under bridges, or for child detection by person height estimation, such as in the case of limiting access to prohibited/dangerous areas.

(18) Referring to FIG. 1A, an image **100** of an object **102** is captured by a camera **103** to be calibrated. Image **100** is bound by image plane **104**. An x-y-z axis reference of a real world coordinate system (referred to as a world plane) is shown in FIG. 1A. Image plane **104** and ground plane **106** may be any plane in the x-y-z space. For simplicity, suppose that image plane **104** is parallel to the x-z plane and ground plane **106** is parallel to the x-y plane. It should be noted that depending on the position of the camera **103** installed in the environment, the tilt of the camera **103**, the curvature of the floor, etc., the representation of image plane **104** and ground plane **106** may be not be as simplistic as described. Nevertheless, the algorithms and execution of the systems and methods in the present disclosure still apply to complicated representations of image plane **104** and ground plane **106**.

(19) In some aspects, calibration parameters include intrinsic parameters (e.g., focal length), which do not depend on the camera position relative to the real world coordinate system, and extrinsic parameters which describe the orientation and location of the camera **103** (e.g., rotation and translation of the camera **103** with respect to some real world coordinate system).

(20) To estimate the calibration parameters, 3-D world points and their corresponding 2-D image points are utilized. To get these correspondences, multiple images of a calibration pattern can be used. By the correspondences, the calibration parameters can be determined.

(21) Regarding the correspondences, a planar homography matrix relates the transformation between two planes (up to a scale factor). For example, the transformation between some world plane to the image plane. To convert between a point on image plane **104** and ground plane **106**, a homography matrix **H** is utilized. The homography matrix provides partial information about the

calibration parameters.

(22) A simple process to find the homography matrix is by matching at least four unique 3-D world points on a real world plane with their corresponding 2-D image points on the image plane. For example, if the homography matrix for an image-to-world ground plane is needed, the chosen points should be on the ground, and homography enables calculating the world location of all the image points on the ground. Given the homography matrix that converts points from an image plane to a ground plane, and two images of an object with known height, the present disclosure describes calibrating a camera **103** and using the calibration to estimate sizes of arbitrary objects captured by the calibrated camera **103**.

(23) FIG. 2 is a diagram **200** showing points on an image plane being mapped to points on a ground plane using a homography matrix. For example, point $(x_{sub.i1}, y_{sub.i1})$ may represent the top most point on object **102** and point $(x_{sub.i2}, y_{sub.i2})$ may represent the bottom most point (aligned with the top most point along the z-axis) on object **102**. These points on image plane **104** represent the perspective of a camera, which does not have a sense of depth. Homography matrix H may be used to convert point $(x_{sub.i1}, y_{sub.i1})$ to point $(x_{sub.g1}, y_{sub.g1})$ and point $(x_{sub.i2}, y_{sub.i2})$ to point $(x_{sub.g2}, y_{sub.g2})$ on ground plane **106**. Points $(x_{sub.g1}, y_{sub.g1})$ and $(x_{sub.g2}, y_{sub.g2})$ are positions of the image plane points in a real world plane. However, it is evident that even if $(x_{sub.i2}, y_{sub.i2})$ is a point on the ground and is accurately positioned as $(x_{sub.g2}, y_{sub.g2})$ on ground plane **106**, the height associated with point $(x_{sub.i1}, y_{sub.i1})$ is not represented by $(x_{sub.g1}, y_{sub.g1})$. Point $(x_{sub.i1}, y_{sub.i1})$ should also be mapped to $(x_{sub.g2}, y_{sub.g2})$ because the top point of object **102** is positioned on the same point as the bottom point of object **102**. However, the height properties of object **102** cause the mapping to be to $(x_{sub.g1}, y_{sub.g1})$.

(24) In the first approach, image to ground homography and two images of an object of known size are used to calibrate the camera **103**, which can then be used to estimate real world distances on the vertical axis (e.g., object height) in an image. In this approach, a calibration component installed on a computing device (e.g., the camera **103** itself or a server connected to the camera **103**) is configured to obtain a first image and estimate two world points. The first is $a_{sub.1} = (x_{sub.1}, y_{sub.1}, 0)$, where $a_{sub.1}$ is the bottom-most point on the object (e.g., the feet location of a person). Point $a_{sub.1}$ corresponds to point $(x_{sub.g2}, y_{sub.g2})$ in FIG. 2. The calibration component estimates $a_{sub.1}$ using the homography matrix H and, for example, an object bottom-most point, (e.g., a foot or feet position in a case of the object being a person) in pixels on the image plane. The second point is $p_{sub.1}$, which is the intersection of the ground and a ray $r_{sub.1}$ extending from the camera center to the top-most point of the object (e.g., a top of a head of a person). The calibration component calculates $p_{sub.1}$ using the homography matrix H and, for example, the head position in pixels on the image plane. Point $p_{sub.1}$ corresponds to point $(x_{sub.g1}, y_{sub.g1})$ in FIG. 2.

(25) Referring to FIG. 3A, for example, graph **300** includes a first vector that intersects with a point on the object and a ground plane. Graph **300** is captured along the y-z plane in the world plane of FIGS. 1A and 1B and depicts point $a_{sub.1}$ and $p_{sub.1}$. It should be noted that during the calibration phase height “ h ” of the object is known. Accordingly, the top-most point on the object is $(x_{sub.1}, y_{sub.1}, h)$.

(26) The calibration component may then obtain a second image from the camera **103**. FIG. 1B is a second image **150** of the object **102** on a different location on the ground plane. Again, the calibration component determines two world points. The first world point is $a_{sub.2} = (x_{sub.2}, y_{sub.2}, 0)$, which is the bottom-most point on the object. The second point is $p_{sub.2}$, which is the intersection of the ground and a ray $r_{sub.2}$ extending from the camera center to the top-most point of the object (e.g., the head of the person). In this case, the top-most point of the object is $(x_{sub.2}, y_{sub.2}, h)$. The calibration component then determines an intersection of $r_{sub.1}$ and $r_{sub.2}$, which serves as the camera center C in the world frame or world coordinate system (i.e., the height of the center of camera **103** in the real world relative to the ground plane) by estimating ray $r_{sub.1}$ by

extending from the point p.sub.1 to (x.sub.1, y.sub.1, h) and r.sub.2 by extending the point p.sub.2 to (x.sub.2, y.sub.2, h).

(27) Suppose that x.sub.2=3 ft, and y.sub.2=4 ft. Accordingly, a.sub.2=(3, 4, 0). Suppose that the coordinates of p.sub.2 are (3, 8, 0). This suggests that the length difference along the y-axis between a.sub.2 and p.sub.2 is 4 feet. Suppose that the known height of the object is 6 feet. The point above a.sub.2 is thus (3, 4, 6). Given these two points, a first ray r.sub.2 extending to camera **103** may be determined using the line equation $z=-1.5y+12$. Suppose that x.sub.1=3 ft, and y.sub.1=10 ft. The coordinates of the point above a.sub.1 is thus (3, 10, 6). Suppose that the coordinates of p.sub.1 are (3, 20, 0). Given these two points, a second ray r.sub.1 extending to camera **103** may be determined using the line equation $z=-0.6y+12$. It should be noted that these are simplistic line equations in which it is assumed that the object simply moves in a straight line towards the camera and does not shift sideways. Both of these rays intersect at point C, which is given by (0, 12). This suggests that the camera is fixed in a spot 12 ft above the ground plane.

(28) Referring to FIG. 3B, for example, graph **350** includes an intersection between the first vector (i.e., r.sub.1) and a second vector (i.e., r.sub.2) drawn based on the object at a different position. The determination of C serves as the calibration of the camera **103**.

(29) Subsequently, when a new object with an unknown height is captured by the calibrated camera **103** in a single input image, the calibration component estimates a ground location a.sub.3=(x.sub.3, y.sub.3, 0) of the object using the homography matrix (e.g., converting the pixel location of the feet of a person to the real world location of the feet). The calibration component then finds, using the homography matrix H, an intersection p.sub.3 between the ground plane and a ray r.sub.3 that extends from point C to the top-most point of the object (e.g., the head of the person). The calibration component then estimates the height of the object given its position on the ground plane. For example, given the height of the camera, at least one point is known. Given p.sub.3, there are two known points and a ray r.sub.3 can be formed. If the y position of a.sub.3 is then entered in the line equation of r.sub.3, the value of h can be determined. For example, if p.sub.3 is (3, 7, 0) and the camera is positioned on (3, 0, 12), the line equation is $z=-(12/7)y+12$. If a.sub.3 is (3, 3, 0), then $z=-(12/7)(3)+12=6.86$. This suggests that the height is approximately 6.86 of the arbitrary object.

(30) On a high level, it should be noted that $r.sub.3=p.sub.3+\lambda(p.sub.3-C)$. To find the height of the object, the calibration components finds $\lambda.sub.z$ for which $r.sub.z=z.sub.3$ (i.e., $\lambda.sub.z=(z.sub.3-p.sub.z)/(p.sub.z-C.sub.z)$). The calibration component determines the height on the point where $\lambda=\lambda.sub.z$. More specifically, $height=C.sub.z((z.sub.3-p.sub.z)/(C.sub.z-p.sub.z))$.

(31) Referring to FIG. 4, in an example of the second approach for calibration and subsequent measurement using parallel lines on the ground in an image, along with a camera focal length and an object of known height in the image, a graph **400** including an image frame **402** having an image of object **404** (e.g., a cabinet) on a surface with parallel lines is captured by a camera **103** to be calibrated. It should be noted that parallel lines **408** and parallel lines **406** are parallel in the real world. However, due to factors such as the curvature/shape of the lens of the camera **103** and the angle at which the camera **103** is positioned, parallel lines in the real world appear angled in the image. For example, in graph **400** and image frame **402**, parallel lines **406** are angled inward such that they would intersect from the perspective of the camera **103**. Likewise, parallel lines **408** are angled inward such that they too would intersect.

(32) Given two sets of parallel lines (e.g., parallel lines **408** and parallel lines **406**), wherein one set intersects with the other set (e.g., are perpendicular or cross at an angle), a camera focal length f, and an image of an object with a known height l.sub.0, the calibration component can calibrate the camera **103** and estimate the heights of arbitrary objects. More specifically, the calibration component determines an intersection point at ∞ of parallel lines **406**, namely, p.sub.1, ∞ , and the intersection point of parallel lines **408**, namely, p.sub.2, ∞ . It should be noted that points p.sub.1, ∞ and p.sub.2, ∞ extend beyond image frame **202**. The calibration component may generate a graph

where each pixel of image frame **402** represents a unit on the graph (as shown in FIG. **4**). In some aspects, the bottom left pixel of image frame **402** serves as the origin point of the graph (0,0). The graph may extend as far as needed depending on where the parallel lines intersect.

(33) The calibration component then determines the projection of the ground line at ∞ , namely, infinity line **410** (used interchangeably with $l_{sub.g,\infty}$ and ground line **510**). Using the focal length f of the camera **103** and the image size in pixels of image frame **402**, the calibration component determines an internal camera matrix K . The calibration component then uses K and infinity line **410** to find any point at infinity, $m_{sub.\infty}$ on any line perpendicular to the ground line. The determination of $m_{sub.\infty}$ serves as the calibration of the camera **103**.

(34) Referring to FIG. **5**, diagram **500** includes a detection plane **506** associated with the camera **103** calibrated based on the graph **400** and image frame **402** of FIG. **4**. In diagram **500**, the height of object **502** is known and the calibration component determines the unknown height of object **504** using the calibrated camera **103**. In particular, when a new object enters a scene in an image frame captured by camera **103**, the calibration component determines the intersection $p_{sub.\infty}$

(intersection **512**) of (A) the line connecting the bottom-most point of the new object (e.g., the feet of object **504**) and the bottom-most point of the known object (e.g., the feet of object **502**) and (B) ground line **510** (e.g., infinity line **410** from FIG. **4**). In this case, object **502** is a stationary object that is used as reference for comparison with other objects.

(35) The calibration component further calculates a virtual line **514** at intersection **512**. Virtual line **514** is perpendicular to ground line **510** and passes through intersection **512**. From virtual line **514**, the calibration component determines detection plane **506**, which is a plane including at least one point on each of the two objects and virtual line **514**. As shown in FIG. **5**, which shows detection plane **506** partially shaded (particularly where the two objects are located). It should be noted that detection plane **506** extends to encompass virtual line **514**. The calibration component determines the homography matrix that transform virtual line **514** to its canonical form $(0, 0, 1)_{sup.T}$ and uses the homography matrix to rectify the top-most and bottom-most points of the objects **502** and **504**. For example, the calibration component uses the feet and head points of the persons in diagram **500** (e.g., $f_{sub.i}$, $h_{sub.i}$, $i=1, 2$, respectively). The calibration component then calculates the rectified heights $l_{sub.r,i}=|h_{sub.i}-f_{sub.i}|$, where the height of object **504** is given by $l_{sub.0}(l_{sub.r,1})/h_{sub.r,2}$.

(36) To better explain this calculation, consider the following example:

(37) In FIG. **5**, suppose that ground line **510** is represented by $\{\overrightarrow{(t_{sub.g,\infty})}\}$ and is given by $\{\overrightarrow{(t_{sub.g,\infty})}\}=(3/9, 1, -320)$. Suppose that vector **508** is represented by $\{\overrightarrow{(t_{sub.g,\infty})}\}$ connecting the feet of object **504** with the feet of object **502**, and is given by $\{\overrightarrow{(t_{sub.f2f})}\}=(-0.38, 1, 147)$. Intersection **512** is the intersection of $\{\overrightarrow{(t_{sub.g,\infty})}\}$ and $\{\overrightarrow{(t_{sub.f2f})}\}$, and is given by $\{\overrightarrow{(t_{sub.f2f})}\} \times \{\overrightarrow{(t_{sub.g,\infty})}\}=(243, 239, 1)$. This point is the first point on the line at infinity of detection plane **506** (e.g., plane II). Suppose that the two lines $\{\overrightarrow{(l_{sub.1})}\}$ and $\{\overrightarrow{(l_{sub.2})}\}$ are given by $\{\overrightarrow{(l_{sub.1})}\}=(78.5, 1, -54903)$ and $\{\overrightarrow{(l_{sub.2})}\}=(-71, 1, 37420)$. These two lines intersect at the point $\{\overrightarrow{(m_{sub.\infty})}\}$ (not shown in the figure) given by $\{\overrightarrow{(m_{sub.\infty})}\}=(617, 642, 1)$. Virtual line **514** is represented by $\{\overrightarrow{(t_{sub.\pi,\infty})}\}$ and is the line between intersection **512** and $\{\overrightarrow{(m_{sub.\infty})}\}$ and is given by $\{\overrightarrow{(t_{sub.\pi,\infty})}\}=(-4.4 \text{ e-}3, 2.6 \text{ e-}4, 1)$. The homography matrix H that rectifies this line by to the form $(0, 0, 1)$ is given by

$$(38) H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -6186 & 374 & 1413583 \end{pmatrix}.$$

(39) Calibration component **615** uses H to rectify the head and feet points of object **502** and **504**. The rectified head and feet points of object **504** are $h_{sub.2}=(-2.5 \text{ e-}4, -9.6 \text{ e-}5, 1)$, $f_{sub.2}=(-2.6$

e-4, -1.6 e-4, 1) and the rectified head and feet points of object **502** are h.sub.1=(-3 e-4, -1.2 e-4, 1), f.sub.1=(-3 e-4, -2 e-4, 1). After the projective rectification, the length ratios are the conserved. Accordingly, if the object **502** height (e.g., h.sub.object) is known to be 2 meters, then:

$$(40) \frac{\text{.Math. } h_1 - f_1 \text{ .Math.}}{\text{.Math. } h_2 - f_2 \text{ .Math.}} = \frac{2}{h_{\text{person}}} \text{ .fwdarw. } h_{\text{person}} = \frac{\text{.Math. } h_2 - f_2 \text{ .Math.}}{\text{.Math. } h_1 - f_1 \text{ .Math.}} * 2 = 0.1 * 2 = 0.2m.$$

(41) Referring to FIG. **6** and FIG. **7**, in operation, computing device **600** may perform a method **700** of estimating a real world height/size in an image, by such as via execution of calibration component **615** by processor(s) **605** and/or one or more memories **610**. For example, method **700** and computing device **900** relate to the first approach that includes using image to ground homography and two images of an object of known size to calibrate the camera **103**, which can then be used to estimate real world heights and sizes in an image.

(42) At block **702**, the method **700** includes obtaining a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or obtaining component **620** may be configured to or may comprise means for obtaining a first image (e.g., image **100**) from a camera **103** installed at an unknown height in an environment, wherein the first image includes a first object (e.g., object **102**) at a first location in the environment, wherein the first object has a known object height (e.g., 6 feet). In some aspects, calibration component **615** identifies the first object by detecting (e.g., using a person detection classifier) an object bounding box around the first object.

(43) At block **704**, the method **700** includes estimating, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or estimating component **625** may be configured to or may comprise means for estimating, using a homography matrix that transforms an image plane (e.g., image plane **104**) associated with the image (e.g., image **100**) to a real world plane, a first position (e.g., a.sub.1=(x.sub.1, y.sub.1, 0)) of a first point (e.g., the feet) on the first object in the first image based on a real world ground plane (e.g., ground plane **106**).

(44) At block **706**, the method **700** includes determining, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or determining component **630** may be configured to or may comprise means for determining, based on the known object height, a first vector (e.g., r.sub.1) that intersects with a second point on the first object (e.g., the head of a person) and points to the real world ground plane (e.g., intersecting at p.sub.1).

(45) At block **708**, the method **700** includes obtaining a second image from the camera, wherein the second image includes the first object at a second location in the environment. In some aspects, the first object in the second image is also bound by an object bounding box, which is detected by calibration component **615**. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or obtaining component **635** may be configured to or may comprise means for obtaining a second image from the camera **103**, wherein the second image includes the first object (e.g., object **102**) at a second location in the environment.

(46) At block **710**, the method **700** includes estimating, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or estimating component **640** may be configured to or may comprise means for estimating, using the homography matrix, a second position (e.g., a.sub.2=(x.sub.2, y.sub.2, 0)) of the first point (e.g., the feet) on the first object in the second image based

on the real world ground plane (e.g., ground plane **106**).

(47) At block **712**, the method **700** includes determining, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or determining component **645** may be configured to or may comprise means for determining, based on the known object height, a second vector (e.g., r.sub.2) that intersects with the second point (e.g., the head) on the first object in the second image and points to the real world ground plane.

(48) At block **714**, the method **700** includes calibrating the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or calibrating component **650** may be configured to or may comprise means for calibrating the camera **103** by determining a height of the center of a camera lens of camera **103** (e.g., C) using an intersection of the first vector (e.g., r.sub.1) and the second vector (e.g., r.sub.2). In some aspects, the exact three-dimensional location of the camera lens center is determined. This location includes the height and lateral position of the camera lens center.

(49) In an alternative or additional aspect, the second point on the first object has a known second point height based on the known object height.

(50) In an alternative or additional aspect, the first point is associated with a base of the first object (e.g., the bottom-most point), and the second point is associated with a top of the first object (e.g., the top-most point).

(51) In an alternative or additional aspect, the first object is a person, the first point is associated with a foot of the person, and the second point is associated with a head of the person.

(52) Referring to FIG. **8**, in an alternative or additional aspect, at block **802**, the method **700** may further include receiving a third image from the camera, wherein the third image includes a second object without a known height in the environment. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or obtaining component **655** may be configured to or may comprise means for obtaining a third image from the camera **103**, wherein the third image includes a second object without a known height in the environment. For example, the third image may be similar to image **100**, but the object in the third image may be different from object **102**. It should be noted that although image **100** includes a person as the object used to calibrate the camera **103**, once the camera **103** is calibrated, the object may be any item, person, animal, etc. The object may be animate or inanimate and does not need to be the same type of object used to calibrate the camera **103**. The object in the third image can also be in any location of the environment as long as the base of the object and the top of the object is visible in the third image—to calculate the size (e.g., height) of the object.

(53) In this optional aspect, at block **804**, the method **700** may further include determining a height metric of the second object using the homography matrix and the height of the center of the camera lens. For example, in an aspect, computing device **600**, processor(s) **605**, one or more memories **610**, calibration component **615**, and/or determining component **660** may be configured to or may comprise means for determining a height metric of the second object using the homography matrix and the height of the center of the camera lens of camera **103** (e.g., C).

(54) In this optional aspect, at block **806**, the determining at block **804** of the height metric of the second object comprises estimating, using the homography matrix, a third position (e.g., a.sub.3=(x.sub.3, y.sub.3, 0)) of a third point (e.g., the feet) on the second object based on the real world ground plane (e.g., ground plane **106**).

(55) In this optional aspect, at block **808**, the determining at block **804** of the height metric of the second object comprises determining, based on the height of the center of the camera lens, a third vector (e.g., r.sub.3) that intersects with a fourth point (e.g. the head) on the second object and

points to the ground plane (e.g., intersecting at p.sub.3).

(56) In this optional aspect, at block **810**, the determining at block **804** of the height metric of the second object comprises determining the height metric between the third point and the fourth point on the second object based on a distance between the third point and the fourth point relative to the height of the center of the camera lens of camera **103** (e.g., C).

(57) For example, calibration component **615** may calculate the height as equal to $C.sub.x((x.sub.3-p.sub.x)/(C.sub.x-p.sub.x))$, as discussed previously.

(58) In an alternative or additional aspect, the third point is associated with a base of the second object, and the fourth point is associated with a top of the second object.

(59) Referring to FIG. **9** and FIG. **10**, in operation, computing device **900** may perform a method **1000** of estimating a real world distance in an image, by such as via execution of calibration component **915** by processor(s) **905** and/or one or more memories **910**. For example, method **700** and computing device **900** relate to the second approach for calibration and subsequent measurement using parallel lines on the ground in an image, along with a camera focal length and an object of known height in the image.

(60) At block **1002**, the method **1000** includes obtaining an image from a camera installed in an environment. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or obtaining component **920** may be configured to or may comprise means for obtaining an image (e.g., image frame **402**) from a camera **103** installed in an environment.

(61) At block **1004**, the method **1000** includes identifying, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or identifying component **925** may be configured to or may comprise means for identifying, in the image, a first set of parallel lines (e.g., parallel lines **406**) on a real world ground plane and a second set of parallel lines (e.g., parallel lines **408**) on the real world ground plane.

(62) At block **1006**, the method **1000** includes determining a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or determining component **930** may be configured to or may comprise means for determining a first intersection point (e.g., p.sub.1,∞) of the first set of parallel lines and a second intersection point of the second set of parallel lines (e.g., p.sub.2,∞).

(63) At block **1008**, the method **1000** includes identifying a ground line connecting the first intersection point and the second intersection point. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or identifying component **935** may be configured to or may comprise means for identifying a ground line (e.g., infinity line **410**) connecting the first intersection point and the second intersection point.

(64) At block **1010**, the method **1000** includes estimating an internal camera matrix based on a focal length and an image size associated with the camera. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or estimating component **940** may be configured to or may comprise means for estimating an internal camera matrix (e.g., K) based on a focal length (e.g., f) and an image size (e.g., of image frame **402**) associated with the camera **103**.

(65) At block **1012**, the method **1000** includes calibrating the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the internal camera matrix. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or calibrating component **945** may be configured to or may comprise means for calibrating the camera **103** for real world distance measurement by determining a perpendicular vector (e.g., m.sub.∞) to the ground line based on the internal camera

matrix.

(66) Referring to FIG. **11**, in an alternative or additional aspect wherein the image or a subsequent image from the camera of the environment includes a first object having a known height and a second object having an unknown height, at block **1102**, the method **1000** may further include determining a height metric of the second object using the perpendicular vector to the ground line and the known height of the first object. For example, in an aspect, computing device **900**, processor(s) **905**, one or more memories **910**, calibration component **915**, and/or determining component **950** may be configured to or may comprise means for determining a height metric of the second object (e.g., object **504**) using the perpendicular vector (e.g., m.sub.∞) to the ground line (e.g., infinity line **410**) and the known height of the first object (e.g., object **502**).

(67) In this optional aspect, at block **1104**, the determining at block **1102** of the height metric of the second object comprises determining an intersection between the ground line (e.g., infinity line **410**) and a vector (e.g., vector **508**) through a first ground point on the first object and a second ground point on the second object.

(68) In this optional aspect, at block **1106**, the determining at block **1102** of the height metric of the second object comprises calculating a virtual line (e.g., virtual line **514**) in a plane (e.g., detection plane **506**) that comprises the first object (e.g., object **502**) and the second object (e.g., object **504**) that is perpendicular to the ground line (e.g., ground line **510**) and includes the intersection (e.g., intersection **512**).

(69) In this optional aspect, at block **1108**, the determining at block **1102** of the height metric of the second object comprises determining a homography matrix that transforms the virtual line to a canonical form. For example, calibration component **915** determines the homography matrix that transforms virtual line **514** to its canonical form. For example, given the coordinates of the point A(x.sub.0, y.sub.0) and the coordinates of the point n={l; m}, the equation of the line can be written in the canonical form using the following formula:

$$(70) \frac{x-x_0}{l} = \frac{y-y_0}{m}$$

(71) In this optional aspect, at block **1110**, the determining at block **1102** of the height metric of the second object comprises calculating a first rectified top point on the first object and second rectified top point on the second object using the homography matrix. For example, calibration component **915** uses the feet and head points of the persons in diagram **500** (e.g., f.sub.i, h.sub.i, i=1, 2, respectively).

(72) In this optional aspect, at block **1112**, the determining at block **1102** of the height metric of the second object comprises calculating the height metric of the second object based on the first rectified top point and the first ground point of the first object and the second rectified top point and the second ground point of the second object. Calibration component **915** then calculates the rectified heights l.sub.r,i=|h.sub.i-f.sub.i|, where the height of object **504** is given by l.sub.0(l.sub.r,1)/h.sub.r,2. Here, the height l.sub.0 is known.

(73) The following clauses include additional implementations of the present disclosure.

(74) Clause 1. A method for estimating a real world distance in an image, comprising: obtaining a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height; estimating, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane; determining, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane; obtaining a second image from the camera, wherein the second image includes the first object at a second location in the environment; estimating, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane; determining, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane;

and calibrating the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

(75) Clause 2. The method of clause 1, wherein the second point on the first object has a known second point height based on the known object height.

(76) Clause 3. The method of any one or any combination of the preceding clauses, wherein the first point is associated with a base of the first object, and the second point is associated with a top of the first object.

(77) Clause 4. The method of any one or any combination of the preceding clauses, wherein the first object is a person, the first point is associated with a foot of the person, and the second point is associated with a head of the person.

(78) Clause 5. The method of any one or any combination of the preceding clauses, further comprising: obtaining a third image from the camera, wherein the third image includes a second object without a known height in the environment; and determining a height metric of the second object using the homography matrix and the height of the center of the camera lens of the camera.

(79) Clause 6. The method of any one or any combination of the preceding clauses, wherein determining the height metric of the second object comprises: estimating, using the homography matrix, a third position of a third point on the second object based on the real world ground plane; determining, based on the height of the center of the camera lens, a third vector that intersects with a fourth point on the second object and points to the ground plane; and determining the height metric between the third point and the fourth point on the second object based on a distance between the third point and the fourth point relative to the height of the center of the camera lens of the camera.

(80) Clause 7. The method of any one or any combination of the preceding clauses, wherein the third point is associated with a base of the second object, and the fourth point is associated with a top of the second object.

(81) Clause 8. An apparatus for estimating a real world distance in an image, comprising: a memory; and a processor coupled with the memory and configured to: obtain a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height; estimate, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane; determine, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane; obtain a second image from the camera, wherein the second image includes the first object at a second location in the environment; estimate, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane; determine, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane; and calibrate the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

(82) Clause 9. The apparatus of clause 8, wherein the second point on the first object has a known second point height based on the known object height.

(83) Clause 10. The apparatus of any one or any combination of the preceding clauses, wherein the first point is associated with a base of the first object, and the second point is associated with a top of the first object.

(84) Clause 11. The apparatus of any one or any combination of the preceding clauses, wherein the first object is a person, the first point is associated with a foot of the person, and the second point is associated with a head of the person.

(85) Clause 12. The apparatus of any one or any combination of the preceding clauses, wherein the one or more processors individually or in combination are further configured to: obtain a third

image from the camera, wherein the third image includes a second object without a known height in the environment; and determine a height metric of the second object using the homography matrix and the height of the center of the camera lens.

(86) Clause 13. The apparatus of any one or any combination of the preceding clauses, wherein to determine the height metric of the second object comprises to: estimate, using the homography matrix, a third position of a third point on the second object based on the real world ground plane; determine, based on the height of the center of the camera lens, a third vector that intersects with a fourth point on the second object and points to the ground plane; and determine the height metric between the third point and the fourth point on the second object based on a distance between the third point and the fourth point relative to the height of the center of the camera lens.

(87) Clause 14. The apparatus of any one or any combination of the preceding clauses, wherein the third point is associated with a base of the second object, and the fourth point is associated with a top of the second object.

(88) Clause 15. A method for estimating a real world distance in an image, comprising: obtaining an image from a camera installed in an environment; identifying, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane; determining a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines; identifying a ground line connecting the first intersection point and the second intersection point; estimating an internal camera matrix based on a focal length and an image size associated with the camera; and calibrating the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the internal camera matrix.

(89) Clause 16. The method of clause 15, wherein the image or a subsequent image from the camera of the environment includes a first object having a known height and a second object having an unknown height, further comprising: determining a height metric of the second object using the perpendicular vector to the ground line and the known height of the first object.

(90) Clause 17. The method of any one or any combination of the preceding clauses, wherein determining the height metric of the second object comprises: determining an intersection between the ground line and a vector through a first ground point on the first object and a second ground point on the second object; calculating a virtual line in a plane comprising the first object and the second object that is perpendicular to the ground line and includes the intersection; determining a homography matrix that transforms the virtual line to a canonical form; calculating a first rectified top point on the first object and second rectified top point on the second object using the homography matrix; and calculating the height metric of the second object based on the first rectified top point and the first ground point of the first object and the second rectified top point and the second ground point of the second object.

(91) Clause 18. An apparatus for estimating a real world distance in an image, comprising: a memory; and a processor coupled with the memory and configured to: obtain an image from a camera installed in an environment; identify, in the image, a first set of parallel lines on a real world ground plane and a second set of parallel lines on the real world ground plane; determine a first intersection point of the first set of parallel lines and a second intersection point of the second set of parallel lines; identify a ground line connecting the first intersection point and the second intersection point; estimate an internal camera matrix based on a focal length and an image size associated with the camera; and calibrate the camera for real world distance measurement by determining a perpendicular vector to the ground line based on the internal camera matrix.

(92) Clause 19. The apparatus of clause 18, wherein the image or a subsequent image from the camera of the environment includes a first object having a known height and a second object having an unknown height, wherein the one or more processors individually or in combination are further configured to: determine a height metric of the second object using the perpendicular vector to the ground line and the known height of the first object.

- (93) Clause 20. The apparatus of any one or any combination of the preceding clauses, wherein to determine the height metric of the second object comprises to: determine an intersection between the ground line and a vector through a first ground point on the first object and a second ground point on the second object; calculate a virtual line in a plane comprising the first object and the second object that is perpendicular to the ground line and includes the intersection; determine a homography matrix that transforms the virtual line to a canonical form; calculate a first rectified top point on the first object and second rectified top point on the second object using the homography matrix; and calculate the height metric of the second object based on the first rectified top point and the first ground point of the first object and the second rectified top point and the second ground point of the second object.
- (94) Clause 21. The apparatus of any one or any combination of the preceding clauses, wherein the height metric of the second object is proportional to the known height of the first object.
- (95) Clause 22. The apparatus of any one or any combination of the preceding clauses, wherein the first object is a stationary object in the environment.
- (96) Clause 23. The apparatus of any one or any combination of the preceding clauses, wherein the first set of parallel lines intersect with the second set of parallel lines.
- (97) Clause 24. The apparatus of any one or any combination of the preceding clauses, wherein the first set of parallel lines are perpendicular to the second set of parallel lines in the real world ground plane.
- (98) While the foregoing disclosure discusses illustrative aspects and/or embodiments, it should be noted that various changes and modifications could be made herein without departing from the scope of the described aspects and/or embodiments as defined by the appended claims. Furthermore, although elements of the described aspects and/or embodiments may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Additionally, all or a portion of any aspect and/or embodiment may be utilized with all or a portion of any other aspect and/or embodiment, unless stated otherwise.

Claims

1. A method for estimating a real world height in an image, comprising: obtaining a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height; estimating, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane; determining, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane; obtaining a second image from the camera, wherein the second image includes the first object at a second location in the environment; estimating, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane; determining, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane; and calibrating the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.
2. The method of claim 1, wherein the second point on the first object has a known second point height based on the known object height.
3. The method of claim 1, wherein the first point is associated with a base of the first object, and the second point is associated with a top of the first object.
4. The method of claim 1, wherein the first object is a person, the first point is associated with a foot of the person, and the second point is associated with a head of the person.
5. The method of claim 1, further comprising: obtaining a third image from the camera, wherein the

third image includes a second object without a known height in the environment; and determining a height metric of the second object using the homography matrix and the height of the center of the camera lens.

6. The method of claim 5, wherein determining the height metric of the second object comprises: estimating, using the homography matrix, a third position of a third point on the second object based on the real world ground plane; determining, based on the height of the center of the camera lens, a third vector that intersects with a fourth point on the second object and points to the ground plane; and determining the height metric between the third point and the fourth point on the second object based on a distance between the third point and the fourth point relative to the height of the center of the camera lens.

7. The method of claim 6, wherein the third point is associated with a base of the second object, and the fourth point is associated with a top of the second object.

8. An apparatus for estimating a real world height in an image, comprising: one or more memories; and one or more processors coupled with the one or more memories, wherein the one or more processors are configured, individually or in combination, to: obtain a first image from a camera installed at an unknown height in an environment, wherein the first image includes a first object at a first location in the environment, wherein the first object has a known object height; estimate, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane; determine, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane; obtain a second image from the camera, wherein the second image includes the first object at a second location in the environment; estimate, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane; determine, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane; and calibrate the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

9. The apparatus of claim 8, wherein the second point on the first object has a known second point height based on the known object height.

10. The apparatus of claim 8, wherein the first point is associated with a base of the first object, and the second point is associated with a top of the first object.

11. The apparatus of claim 8, wherein the first object is a person, the first point is associated with a foot of the person, and the second point is associated with a head of the person.

12. The apparatus of claim 8, wherein the one or more processors are further configured to: obtain a third image from the camera, wherein the third image includes a second object without a known height in the environment; and determine a height metric of the second object using the homography matrix and the height of the center of the camera lens.

13. The apparatus of claim 12, wherein to determine the height metric of the second object comprises to: estimate, using the homography matrix, a third position of a third point on the second object based on the real world ground plane; determine, based on the height of the center of the camera lens, a third vector that intersects with a fourth point on the second object and points to the ground plane; and determine the height metric between the third point and the fourth point on the second object based on a distance between the third point and the fourth point relative to the height of the center of the camera lens.

14. The apparatus of claim 13, wherein the third point is associated with a base of the second object, and the fourth point is associated with a top of the second object.

15. A non-transitory computer-readable medium storing instructions for estimating a real world height in an image, wherein the instructions are executable by one or more processors to: obtain a first image from a camera installed at an unknown height in an environment, wherein the first

image includes a first object at a first location in the environment, wherein the first object has a known object height; estimate, using a homography matrix that transforms an image plane associated with the image to a real world plane, a first position of a first point on the first object in the first image based on a real world ground plane; determine, based on the known object height, a first vector that intersects with a second point on the first object and points to the real world ground plane; obtain a second image from the camera, wherein the second image includes the first object at a second location in the environment; estimate, using the homography matrix, a second position of the first point on the first object in the second image based on the real world ground plane; determine, based on the known object height, a second vector that intersects with the second point on the first object in the second image and points to the real world ground plane; and calibrate the camera by determining a height of the center of a camera lens of the camera using an intersection of the first vector and the second vector.

16. The non-transitory” computer-readable medium of claim 15, wherein the second point on the first object has a known second point height based on the known object height.

17. The non-transitory” computer-readable medium of claim 15, wherein the first point is associated with a base of the first object, and the second point is associated with a top of the first object.

18. The non-transitory” computer-readable medium of claim 15, wherein the first object is a person, the first point is associated with a foot of the person, and the second point is associated with a head of the person.

19. The non-transitory” computer-readable medium of claim 15, further comprising instructions executable by the one or more processors to: obtain a third image from the camera, wherein the third image includes a second object without a known height in the environment; and determine a height metric of the second object using the homography matrix and the height of the center of the camera lens.

20. The non-transitory” computer-readable medium of claim 19, wherein an instruction to determine the height metric of the second object comprises instructions executable by the one or more processors to: estimate, using the homography matrix, a third position of a third point on the second object based on the real world ground plane; determine, based on the height of the center of the camera lens, a third vector that intersects with a fourth point on the second object and points to the ground plane; and determine the height metric between the third point and the fourth point on the second object based on a distance between the third point and the fourth point relative to the height of the center of the camera lens.
